

A Streaming Geometry Model Is an Episodic Memory: Monocular Multi-Goal Navigation Without a Pose Oracle

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Abstract—In multi-goal image navigation, a robot may be asked to return to a previously seen place after the relevant view has fallen outside its controller’s short context. The episode history still contains the needed visual evidence, but using it requires relating the recalled view to the robot’s current camera. We show that a frozen streaming geometry model already maintains the persistent spatial state needed for this relation. Geometric Episodic Memory (GEM) pairs this state with an indexed RGB history. Given a new ImageGoal, GEM retrieves and geometrically verifies a supporting historical view. This view then serves as a bridge from the goal to the current camera, yielding a current-relative, scale-free goal bearing without direct goal-to-current image matching. GEM adapts the verified recall to each frozen controller’s native goal interface; when verification fails, it abstains and preserves native execution. The same geometry stream also supplies monocular depth for local control. GEM requires no external pose measurements, explicit metric map, or additional training. In paired evaluations with NavDP, ViNT, and NoMaD on shared causal histories in HM3D and MP3D, GEM improves historically supported Revisit success by 60.7–81.0 percentage points while maintaining or improving Novel success. Native execution is preserved in 97.6% of Novel controller–query pairs.

I. INTRODUCTION

Robots deployed in homes and offices are often asked to return to places they have encountered before. What a robot observes while pursuing one goal may therefore become useful evidence for a later one. Yet pretrained goal-conditioned visual navigation policies [1]–[3] retain only a short observation context: once a goal-relevant view leaves that context, the policy can no longer use it for a later goal.

This limitation matters especially in multi-goal ImageNav [4], where goals arrive sequentially without resetting the environment or observation history. For a Novel goal, useful visual evidence may simply not have been observed yet, and exploration can acquire it. A Revisit is different: the episode may already contain a view that supports the goal, but the controller can no longer access it. Exploration may then waste effort rediscovering evidence that already exists in the episode history. The problem is therefore not merely to store past observations, but to make the relevant one actionable when a later goal calls for it.

Prior visual navigation systems reuse past experience through learned episodic memory [5], metric or graph-structured representations [6], [7], or retrieval of past observations [8], while MemoNav [4] and GOAT-Bench [9] study navigation across sequential goals.

This raises a more basic question: what persistent information is actually needed to make a past observation actionable to an already trained goal-conditioned controller? We study this question using only causal RGB observations, without

external pose measurements or an explicit metric map. We use *Revisit* and *Novel* only as evaluation labels for historically supported and low-overlap goals, respectively; the controller never observes these labels at runtime.

Causal streaming geometry models [10]–[12] maintain persistent geometric state that relates observations as new RGB images arrive. For navigation, recalling a past observation is useful only if the robot can relate that observation to where it is now. Streaming geometry maintains this cross-time spatial relation in a common episode-local frame, providing the spatial basis for episodic recall.

Geometric Episodic Memory (GEM) uses this frozen state as episodic memory, without constructing a separate metric map or training a new memory model.

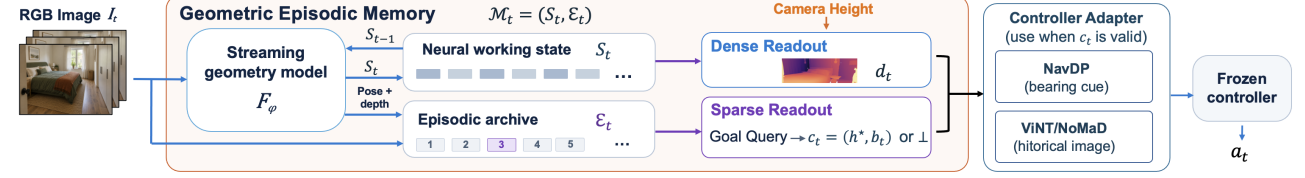
A new ImageGoal, however, is not part of the causal stream and may be captured from an unknown viewpoint. To ground such a goal in memory, GEM associates the streaming geometry with an indexed history of past RGB observations. Given a new ImageGoal, GEM retrieves candidate historical views and geometrically verifies whether any of them supports the goal. A verified historical view then serves as a bridge: it grounds the goal in past visual evidence, while the geometry stream relates that evidence to the current camera. The readout returns the supporting view and a current-relative, scale-free unit bearing, leaving local trajectory generation to the controller.

The same geometry stream also supplies monocular geometry for local control. Goals query this memory without modifying the causal stream, allowing accumulated history and geometry to persist across successive goals. GEM exposes verified recall through the native goal interfaces of NavDP, ViNT, and NoMaD [1]–[3]; when recall cannot be verified, it abstains and preserves the controller’s native request. All model parameters remain frozen; no additional training is required.

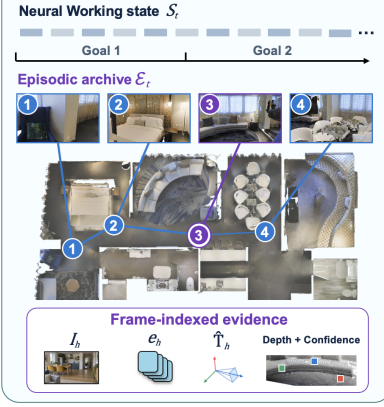
Our contributions are:

- We show that the persistent state of a frozen streaming geometry model can serve as the spatial basis for episodic memory across successive navigation goals.
- We introduce a method that matches a goal image to a historical view, verifies the match geometrically, and uses the shared geometry to estimate the direction from the robot to the goal.
- Across NavDP, ViNT, and NoMaD on HM3D and MP3D, GEM substantially improves Revisit success while preserving Novel performance and native execution when recall is rejected.

(a) System Overview



(b) Episodic Memory



(c) Goal Readout

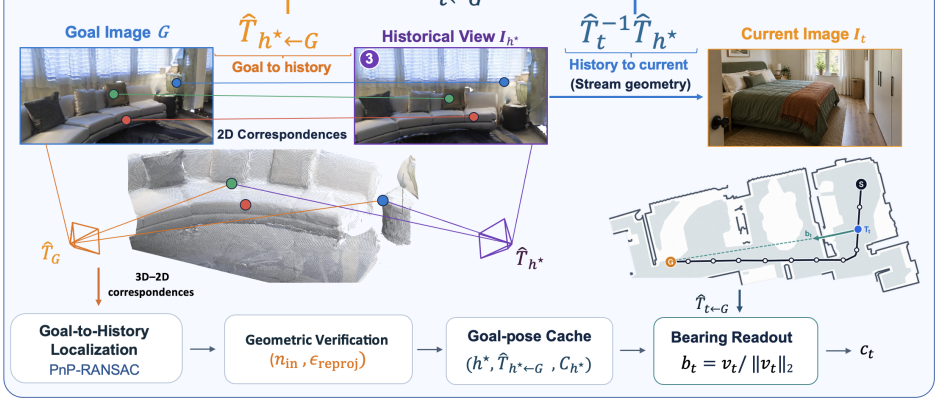


Fig. 1. **Geometric Episodic Memory for frozen controllers.** (a) A frozen streaming geometry model writes observations into GEM. Dense readout supplies NavDP with current depth; sparse readout returns a verified historical view and goal bearing. The adapters give NavDP a bearing cue alongside G ; ViNT/NoMaD receive I_{h^*} . Abstention preserves the native request. (b) The neural working state and episodic archive persist across goals. Each archived view links RGB and a descriptor to its pose, sampled depth, and confidence. Goals query memory without entering the observation stream. (c) Historical 3D–2D correspondences localize the goal. The verified goal-to-history pose fixes the goal in the stream frame; cached localization then supports current-relative bearing updates. Map backgrounds are for illustration.

II. RELATED WORK

Goal-conditioned navigation policies. ViNT learns transferable visual navigation [1]; NoMaD combines goal-conditioned navigation and exploration through goal-masked diffusion [2]. NavDP jointly learns trajectory generation and evaluation from simulation [3]. X-NavDP improves behavioral and embodiment adaptation through reinforcement learning, while retaining short-term temporal context [13]. LoGoPlanner uses geometric history for implicit localization and planning in RGB-D PointGoal navigation, where target coordinates are given [14]. GEM addresses a complementary problem: recovering the direction of an ImageGoal whose location is unknown, so that a frozen controller can navigate to it using past observations.

Memory for visual navigation. Navigation systems store past experience in learned episodic representations [5], semantic maps [6], or graphs of observed places [7], [15], [16]. MemoNav learns goal-relevant working memory over a topological map [4]; AstraNav-Memory compresses visual histories for navigation [17]. GOAT-Bench studies sequential multimodal goals [9]. RANa combines retrieved views with learned geometric context and also supports zero-shot goal replacement [8]. GEM instead reads an explicit, geometrically verified direction from a causal RGB stream and passes it through a frozen controller’s existing interface.

Geometric goal localization. Global descriptors retrieve reference views [18], [19], and local correspondences verify matches and estimate pose [20]. SLING uses current-to-

goal matches for last-mile control [21], while IEVE actively verifies target instances [22]. Gaussian representations support predictive control in BEINGS [23] and incremental goal localization in IGL-Nav [24]. AnyImageNav jointly registers goal and cached views using RGB-D and camera-pose inputs [25]. GEM uses a historical view to localize the goal in geometry already accumulated from the observation stream. The goal queries this geometry without becoming a new observation.

Streaming 3D perception. CUT3R, SStream3R, and LingBot-Map maintain cross-frame geometry from RGB streams [10]–[12]. GEM pairs this working state with a frame-indexed episodic archive, using the shared geometry for both current depth and later goal localization.

III. PROBLEM FORMULATION

Sequential ImageNav. We consider navigation to successive image goals in the same environment [4]. For each active goal G , the robot must reach its capture location within a fixed action budget. At time t , it receives a monocular RGB observation I_t , with prior history $\mathcal{H}_t = (I_0, \dots, I_{t-1})$. Changing goals does not reset the robot’s physical state or accumulated history. Camera intrinsics and mounting height are known; no external pose measurements, metric goal coordinates, or metric map are provided.

Historical support. When a goal G arrives at time t_q , the robot’s history $\mathcal{H}_{t_q}$ may already contain views that overlap the goal image. For evaluation, G is a *Revisit* if at least one historical view meets the covisibility criterion, and *Novel* if the

history meets the low-support criterion (Sec. V). These labels describe historical support for evaluation and are unavailable at runtime.

Memory for a frozen controller. We augment a pretrained goal-conditioned policy π_θ while keeping its parameters fixed. Its native command is $a_t = \pi_\theta(o_t, G)$, where o_t contains the policy’s observation context. Memory $\mathcal{M}_t$ is built causally from RGB through time t . A sparse readout $\mathcal{R}_s(G, \mathcal{M}_t)$ returns a verified record $c_t = (h^*, b_t)$ or abstains with $c_t = \perp$. Here $h^* < t$ identifies the supporting image I_{h^*} , and $b_t \in \mathbb{S}^1$ is a unit goal bearing in the current camera’s forward–left plane. A controller-specific adapter A_π uses this record to set the policy’s goal input and align the robot’s heading when needed:

$$a_t = \begin{cases} \pi_\theta(o_t, G), & c_t = \perp, \\ A_\pi(\pi_\theta, o_t, G, c_t), & c_t \neq \perp. \end{cases} \quad (1)$$

On abstention, the native request and controller state are preserved.

IV. GEOMETRIC EPISODIC MEMORY

GEM makes a past observation usable for a new goal by linking three views: the goal image, a supporting historical view, and the current camera (Fig. 1). The archive provides the historical view; the shared geometry relates it to the camera as the robot moves.

A. Memory State and Causal Writing

Working state and episodic archive. GEM maintains $\mathcal{M}_t = (S_t, \mathcal{E}_t)$. The neural working state S_t provides context for estimating depth and pose as observations arrive. The episodic archive $\mathcal{E}_t$ keeps past images linked to their geometry, so they remain available after leaving the model’s recent-view context. Both components persist across goals and are cleared at episode reset.

Observation-driven writing. After initialization, each RGB observation updates the frozen LingBot-Map [12]:

$$(S_t, \hat{D}_t^{\text{rel}}, \hat{T}_t) = F_\phi(S_{t-1}, I_t). \quad (2)$$

In Eq. 2, $\hat{D}_t^{\text{rel}}$ is depth in the reconstruction’s internal scale, and $\hat{T}_t = (\hat{R}_t, \hat{t}_t) \in \text{SE}(3)$ maps camera t into the common episode-local stream frame. Camera axes are x right, y down, and z forward.

When a frame’s geometry is first estimated, GEM stores the image I_h , its DINOv2-L CLS descriptor $e_h \in \mathbb{R}^{1024}$ [19], pose $\hat{T}_h$, and samples of depth and confidence under the same frame index. The index $\mathcal{H}_t^{\text{idx}} = \{(I_h, e_h) \mid h < t\}$ links each retrieved image to its geometry. Only observations advance the stream; goal images query the stored history.

B. Localizing an ImageGoal from History

Retrieving a historical view. When a new goal G is issued at time t_q , we compare its DINOv2 CLS descriptor with historical descriptors to retrieve a shortlist $\mathcal{Q}_G$ of at most eight views with $8 \leq h < t_q$. Temporal non-maximum suppression keeps shortlisted views at least four frames apart. The shortlist stays fixed until the goal changes.

SuperPoint [26] and LightGlue [27] match each candidate view to G . We rank candidates by fundamental-matrix inliers from MAGSAC [28], using appearance similarity and temporal information to break ties. We estimate the goal pose from the highest-ranked view, h^* , if it has at least 16 epipolar inliers. Only this view is passed to PnP.

Estimating the goal pose. The goal and its supporting view may be captured at different positions. We use the view’s archived geometry to locate the goal camera relative to it. For $h = h^*$, let p_j and q_j denote matched pixels in the historical and goal images, respectively. Using archived depth, we back-project each valid p_j to a 3D point X_j^h in the historical camera frame. The relative pose $\hat{T}_{h \leftarrow G} = (\hat{R}_{h \leftarrow G}, \hat{t}_{h \leftarrow G})$ maps the goal camera frame into the historical camera frame. We project each point into the goal image using the inverse pose:

$$\begin{aligned} X_j^h &= \hat{D}_h^{\text{rel}}(p_j) K_h^{-1} \bar{p}_j, \\ \hat{q}_j &= \text{proj} \left(K_G \hat{R}_{h \leftarrow G}^\top (X_j^h - \hat{t}_{h \leftarrow G}) \right), \end{aligned} \quad (3)$$

where K_h and K_G denote camera intrinsics and $\bar{p}_j$ is a homogeneous pixel coordinate. The operator proj performs perspective division. PnP-RANSAC estimates $\hat{T}_{h \leftarrow G}$ from these 3D–2D correspondences [29], [30]. After verification, we combine this relative pose with the stream poses to express the goal relative to the current camera:

$$\hat{T}_{t \leftarrow G} = \underbrace{\hat{T}_t^{-1} \hat{T}_h}_{\text{history to current}} \underbrace{\hat{T}_{h \leftarrow G}}_{\text{goal to history}}. \quad (4)$$

The historical view provides the visual overlap needed for localization, even when the goal and current images do not overlap.

Verifying geometric support. For the candidate goal pose, let $\mathcal{I}$ denote the PnP inlier set and $n_{\text{in}} = |\mathcal{I}|$. Its per-coordinate reprojection RMSE is

$$\epsilon_{\text{reproj}} = \sqrt{\frac{1}{2n_{\text{in}}} \sum_{j \in \mathcal{I}} \|\hat{q}_j - q_j\|_2^2}. \quad (5)$$

Using Eq. 5, we accept the pose only when

$$C_h = \mathbf{1} \left[\begin{array}{l} \text{valid PnP, } n_{\text{in}} \geq 16, \\ \epsilon_{\text{reproj}} \leq 2.0 \text{ px} \end{array} \right]. \quad (6)$$

C. Sparse and Dense Readouts

Scale-free goal direction. Let u_t denote the translation of $\hat{T}_{t \leftarrow G}$ in Eq. 4. Projecting it onto the camera’s forward–left plane gives

$$v_t = [u_{t,z}, -u_{t,x}]^\top, \quad b_t = \frac{v_t}{\|v_t\|_2}. \quad (7)$$

The sparse readout $\mathcal{R}_s(G, \mathcal{M}_t)$ returns $c_t = (h^*, b_t)$ only if $C_{h^*} = 1$ and v_t is finite with $\|v_t\|_2 > 10^{-12}$; otherwise it returns $\perp$. The accepted pair $(h^*, \hat{T}_{h^* \leftarrow G})$ fixes the goal pose $\hat{T}_{h^*} \hat{T}_{h^* \leftarrow G}$ in the stream frame. We cache this localization or the rejection result until the goal changes. Later reads update the bearing using the current stream pose, without repeating matching or PnP.

Changing the reconstruction by a common rotation $R_0 \in \text{SO}(3)$, translation $\tau \in \mathbb{R}^3$, and positive scale $s > 0$ gives $\tilde{R}_i = R_0 \hat{R}_i$ and $\tilde{t}_i = sR_0 \hat{t}_i + \tau$; the cached relative translation also scales as $\tilde{t}_{h \leftarrow G} = s \hat{t}_{h \leftarrow G}$. Equation 4 therefore gives

$$\tilde{u}_t = s u_t, \quad \tilde{b}_t = \frac{s v_t}{\|s v_t\|_2} = b_t. \quad (8)$$

The unit bearing is independent of the common coordinate frame and uniform scale.

Current depth for local control. NavDP’s depth input is expressed in meters. We calibrate the scale from the ratio of the known mounting height h_{cam} [31] to $\hat{h}_{40}$, the median valid floor-to-camera height in reconstruction units over the first 40 causal observations. The scale $\hat{s}_{40}$ stays fixed for the episode. The dense readout $d_t = \mathcal{R}_d(\mathcal{M}_t)$ is

$$d_t = \hat{D}_t^{\text{mono}} = \begin{cases} 0, & t < 40 \text{ or invalid scale,} \\ \hat{s}_{40} \hat{D}_t^{\text{rel}}, & \text{otherwise.} \end{cases} \quad (9)$$

Each depth map is paired with its RGB frame; the bearing readout uses the unscaled shared geometry.

D. Frozen Controller Interfaces

NavDP is our primary controller and uses both GEM readouts. Given a verified record $c_t = (h^*, b_t)$, it retains the original ImageGoal G and adds a local point cue ρb_t , with fixed radius ρ , through its mixed ImageGoal/PointGoal interface. The point cue supplies a direction for local planning, while G identifies the destination. We also adapt GEM to ViNT and NoMaD [1], [2] by replacing G with the verified historical image I_{h^*} at the policy’s RGB ImageGoal input.

All three adapters interpret b_t in the camera-aligned forward–left control frame. When the goal lies behind the robot, they first turn it in place by a bounded amount. The frozen policy then replans from a fresh RGB observation. As the robot moves, GEM updates the bearing from the cached goal pose and the latest stream geometry.

V. EXPERIMENTS

We test whether GEM makes historical observations useful to frozen controllers, retains that benefit across goals, and provides reliable guidance. NavDP is the primary controller throughout; the main comparison also tests the RGB ImageGoal adapters for ViNT and NoMaD.

A. Experimental Setup

Environments and inputs. We evaluate in Habitat [32] on HM3D [33] and MP3D [34]. The system receives causal RGB, the active ImageGoal, and known camera calibration and mounting height. Simulator geometry supports goal construction, scoring, and collision handling; motion execution uses ideal low-level state. Neither GEM nor the policy receives external pose measurements or a metric map.

Shared histories and query types. For each monocular NavDP history in Table I, we construct one Revisit and one Novel ImageGoal. Base and GEM receive the same RGB history in temporal order before their paired rollouts. Histories contain 96–564 observations (median 189). Initial geodesic

distances to the goal are 2.04–3.93 m for Revisit and 2.25–8.98 m for Novel queries.

Covisibility is the fraction of goal-view surface points that reproject inside a historical view and agree with rendered depth within 0.30 m. Main-table Revisit queries have support in $[0.55, 0.90]$, whereas Novel queries have support below 0.10. Sec. V-F evaluates a broader range of historical support.

Paired controller comparison. We compare each frozen controller (*Base*) with its GEM-augmented counterpart (Sec. IV-D) across six controller–dataset comparisons, totaling 840 query rollouts. Each pair shares the scene, start, goal, history, seed, and control budget. NavDP receives causal monocular depth in both arms; ViNT and NoMaD use RGB only. We run NoMaD in goal-conditioned mode.

Implementation details. The geometry stream starts with eight frames. Matching uses up to 2048 SuperPoint keypoints, pretrained LightGlue, and a 1.5-pixel MAGSAC threshold. PnP uses EPnP with a 3-pixel RANSAC threshold and Levenberg–Marquardt refinement. Depth calibration applies a fixed factor of 1.15 to $h_{\text{cam}}/\hat{h}_{40}$ and clips the scale to $[0.8, 6.0]$. These settings and Eq. 6 are shared across datasets and controllers. Before evaluation, we fix the NavDP cue radius at $\rho = 2.5$ m, rounding the episode-balanced median first-intervention cue length.

Execution and metrics. Each query has a 600-step budget. Controllers replan every eight steps in the main and continuous navigation evaluations and the covisibility and heading studies. Movement and rotation are bounded per step; Habitat handles collisions. Rearward cues trigger turns of at most 4.5° per step, with fresh RGB and all turns counted toward the budget. All methods automatically terminate successfully within 1 m of the original goal-image capture location, measured by planar Euclidean distance. A shared stagnation check can end unsuccessful runs early. We report success and success weighted by path length (SPL) [35].

We use two-sided exact McNemar tests, with separate Holm corrections for the main Revisit comparisons and the planned heading controls. Main-table 95% confidence intervals use 20,000 scene-cluster bootstrap samples.

B. Recall Across Frozen Controllers

GEM improves Revisit success for all three controllers on both datasets (Table I). Success gains span 60.7–81.0 percentage points, with all six paired tests significant after Holm correction ($p_{\text{Holm}} \leq 2.22 \times 10^{-4}$). For NavDP, both arms use the same geometry for current depth, isolating the benefit of historical recall. The ViNT and NoMaD results show that the verified view and bearing can also support controllers with an RGB ImageGoal interface.

Novel success improves for NavDP on HM3D and remains unchanged in the other comparisons. GEM preserves native execution in 205/210 Novel controller–query pairs (97.6%).

Initial bearing accuracy. GEM accepts 69 of the 70 distinct Revisit queries in Table I. Their initial bearing error against ground truth has median 1.72° and 90th percentile 6.63° . For seven queries with 1–2 m between the goal and supporting

TABLE I

REVISIT GAINS ACROSS THREE FROZEN CONTROLLERS. BASE AND GEM SHARE CAUSAL HISTORIES AND FROZEN WEIGHTS. SUCCESS IS GOALS REACHED/QUERIES; REVISIT SUCCESS RATE GAIN (Δ SR) IS IN PERCENTAGE POINTS WITH SCENE-CLUSTER BOOTSTRAP 95% CONFIDENCE INTERVALS.

Dataset	Controller	Revisit success		Revisit SPL		Δ SR [95% CI]	Novel success	
		Base	GEM	Base	GEM		Base	GEM
HM3D	NavDP	10/28	27/28	0.131	0.930	60.7 [38.7, 82.6]	6/28	7/28
	ViNT	5/28	25/28	0.088	0.817	71.4 [51.7, 89.7]	5/28	5/28
	NoMaD	7/28	24/28	0.085	0.751	60.7 [35.7, 81.2]	3/28	3/28
MP3D	NavDP	9/42	41/42	0.090	0.945	76.2 [60.9, 90.0]	13/42	13/42
	ViNT	2/42	36/42	0.027	0.799	81.0 [69.4, 92.1]	6/42	6/42
	NoMaD	8/42	37/42	0.075	0.771	69.0 [51.1, 86.1]	11/42	11/42

HM3D: 28 histories / 21 scenes; MP3D: 42 histories / 25 scenes. NavDP receives RGB and monocular depth; ViNT and NoMaD receive RGB only. Paired Revisit wins/losses by controller: HM3D 17/0, 21/1, 19/2; MP3D 33/1, 34/0, 31/2.

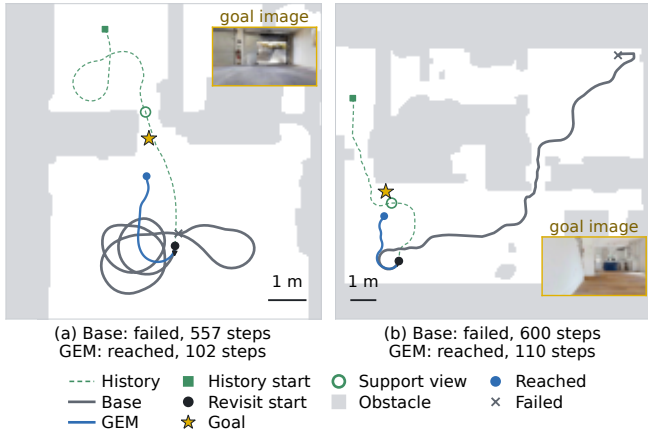


Fig. 2. **Recalling a goal from history.** Paired HM3D Revisit runs share a first-goal history (dashed green) and start at its endpoint. Open green circles mark verified supporting frames; gray/blue paths show Base/GEM. Maps are for visualization.

camera, median error is 1.14° for GEM versus 15.84° when pointing to that camera using ground-truth poses.

Recall beyond recent observations. A post-hoc analysis of Table I identifies 13 queries in 12 scenes with historical support ≥ 0.5 but overlap < 0.1 with the current view and each of the seven preceding decision observations. This window includes the initial context of all three controllers. On these queries, success rises from 3/13 to 11/13 for NavDP, 0/13 to 8/13 for ViNT, and 4/13 to 7/13 for NoMaD.

These queries are also included in the heading and guidance controls (Table IV(b)). Figure 2 shows paired NavDP Revisits.

C. Recall Across Successive Goals

To test memory across goal changes, we use two complementary settings (Table II). In continuous navigation, each method builds its own history. Controlled recall gives both methods the same completed history, isolating third-goal performance from differences in their earlier rollouts.

Continuous three-goal protocol. We evaluate 411 paired $A \rightarrow B \rightarrow C$ runs in 18 MP3D scenes. A is Novel; B and C can be Novel or Revisit. Each method retains its RGB history

TABLE II

NAVDP NAVIGATION ACROSS SUCCESSIVE GOALS. (A) CONTINUOUS THREE-GOAL RUNS WITH EACH METHOD’S OWN HISTORY. (B) THEIR THIRD-GOAL OUTCOMES BY THE TYPE OF B. (C) PAIRED THIRD-GOAL RECALL FROM SHARED TWO-LEG HISTORIES.

(a) Continuous navigation: MP3D, 18 scenes

Stage	Query	Base	GEM
A	Novel	226/411	226/411
	Novel	44/76	43/76
B	Revisit	12/67	63/67
	Novel	7/17	20/40
C	Revisit	5/28	44/44
	Shared Revisit-C	5/28	28/28
A-B-C completed		12	64

Counts are successes/issued queries. C denominators differ because continuation requires success and a valid next goal. Shared Revisit-C compares the same issued goals.

(b) Third-goal outcomes by preceding goal type

Preceding B	Novel-C		Revisit-C	
	Base	GEM	Base	GEM
Novel	6/15	6/15	4/22	22/22
Revisit	1/2	14/25	1/6	22/22

(c) Shared history: HM3D, 20 histories, 13 scenes

Third goal	Success		SPL	
	Base	GEM	Base	GEM
Revisit	8/20	17/20	0.149	0.762
Novel	4/20	4/20	0.176	0.176

and geometric state: B can use observations from A, and C from both A and B.

After each successful goal, we construct the next from the resulting state and history. When both methods remain active, they receive a common goal meeting both histories’ query-type conditions. A method may continue alone if the other stops. Goals are 2–9 m away; Novel goals lie within $\pm 60^\circ$ of the current heading, while Revisit goals have no heading restriction.

Paired recall during continuous navigation. On goals issued to both methods, GEM raises Revisit-B success from 12/67 to 63/67 and Revisit-C from 5/28 to 28/28 (Table II(a)). Novel-B changes from 44/76 to 43/76. GEM reaches every issued Revisit-C goal after either a Novel or a Revisit intermediate goal (Table II(b)).

Chain completion. Completing more Revisits lets GEM continue to C and finish 64 three-goal chains, versus 12 for Base. Another 94 Base and 105 GEM runs stop after a successful prefix because no valid next goal can be constructed.

Controlled third-goal recall. Across 20 shared two-leg histories in 13 HM3D scenes, ten Revisit queries draw support from A and ten from B. GEM raises Revisit success from 8/20 to 17/20 (Table II(c); paired $p = 0.0117$). Novel success remains 4/20 with one intervention.

D. Current Depth and Historical Recall

TABLE III

QUERY-TIME DEPTH AND RECALL ABLATION. 21 REVISIT AND 21 NOVEL QUERIES FROM TEN MP3D SCENES, WITH HISTORIES COLLECTED USING METRIC-DEPTH NAVDP.

Depth input	Recall	Revisit success	Novel success	Overall SPL
Zero input	Off	1/21	3/21	0.060
Simulator	Off	3/21	6/21	0.106
	On	20/21	7/21	0.459
Monocular	Off	6/21	6/21	0.132
	On	20/21	8/21	0.494

Current depth and historical recall serve different purposes. We vary both at query time on 21 fixed histories collected by NavDP with metric depth in ten MP3D scenes (Table III). With recall disabled, NavDP reaches 1/21 Revisit goals with zero depth input, 3/21 with simulator depth, and 6/21 with causal monocular depth. Adding recall raises Revisit success to 20/21 with either nonzero depth source. Historical recall complements current depth by supplying the goal’s direction.

E. Geometric Verification and Control Guidance

Geometric verification. With monocular depth, Table IV(a) compares raw retrieval and PnP-based recall on the queries in Table III. Raw retrieval points toward the camera position of the top-1 retrieved view; PnP only and GEM share top-8 proposals.

Raw retrieval already reaches 20/21 Revisit goals; PnP only and GEM match this result. Verification’s main benefit here is on low-support queries: Novel interventions fall from 21/21 with raw retrieval and 18/21 with PnP only to 2/21 with GEM. Novel success rises from 3/21 and 5/21, respectively, to 8/21. The paired Novel wins/losses are 5/0 against raw retrieval ($p = 0.0625$) and 4/1 against PnP only ($p = 0.375$).

Retrieval supplies a view with at least 0.5 covisibility for 19/21 Revisits at top-1 and all 21 at top-8; geometric ranking selects one for 19/21. The query-label oracle blocks all Novel

TABLE IV

GEOMETRIC VERIFICATION AND CONTROL GUIDANCE. MONOCULAR NAVDP ON (A) TABLE III QUERIES AND (B) 53 TABLE I REVISITS WITH COMPLETE PAIRED RUNS AT THE REPORTING CUTOFF.

(a) Geometric verification				
Memory rule	Revisit		Novel	
	Cue used	Success	Cue used	Success
Raw retrieval (top-1)	21/21	20/21	21/21	3/21
PnP only (top-8)	21/21	20/21	18/21	5/21
GEM (top-8)	21/21	20/21	2/21	8/21
Oracle query label [†]	21/21	20/21	0/21	6/21

(b) Heading alignment and continued guidance	
NavDP configuration	Revisit success
Base	16/53
GEM cues without explicit turns	24/53
Initial GEM alignment, then Base	40/53
Full GEM	51/53

Cue used counts queries with any memory intervention. PnP only checks pose validity; GEM applies Eq. 6. [†] The oracle blocks Novel interventions; it supplies no pose or bearing.

GEM aligns to rearward bearings; Full GEM updates cues and realigns as needed. All turns count toward the 600-step budget.

interventions and reaches 6/21 Novel goals; GEM intervenes on two and reaches 8/21.

Heading alignment and continued guidance. We compare all configurations on the same queries and histories (Table IV(b)). Initial alignment uses the recalled direction and accounts for much of the gain. Continued cues and realignment raise success from 40/53 to 51/53 with Full GEM (12 paired wins, one loss; $p_{\text{Holm}} = 0.0068$). Cues without explicit turns also fall short of Full GEM. These controls show how memory supports both initial orientation and continued guidance.

F. Recall Across Visual Overlap Levels

We vary each goal image’s yaw at a fixed camera position. The 28 monocular histories from 21 HM3D scenes yield 131 queries across overlap levels, plus 28 Novel controls.

GEM improves success in every support bin (Table V), including the lowest-overlap group. Success generally increases with historical support.

G. Computational Cost

Storing geometry for later recall. Local control needs the current depth map, whereas later PnP uses historical depth only at matched keypoints. At write time, we store FP32 depth and confidence in the four-pixel bilinear neighborhoods of frozen SuperPoint keypoints. With the detector and preprocessing fixed, later matches select a subset of these points, giving the same interpolated depths as the full maps.

We profile this archive on a shared RTX 4090. The neural state retains initial views, 64 recent committed views, and historical trajectory tokens. After initialization, a view is committed every seven observations; every observation still produces geometry and enters the archive. With uncompressed

TABLE V

NAVIGATION ACROSS VISUAL OVERLAP LEVELS. NAVDP SUCCESS COUNTS FOR 159 HM3D QUERIES (28 HISTORIES, 21 SCENES). q IS THE MAXIMUM COVISIBILITY OVER ELIGIBLE HISTORICAL FRAMES ($h \geq 8$).

Support q	Base	GEM
[0.1, 0.3)	5/14	9/14
[0.3, 0.5)	9/26	23/26
[0.5, 0.7)	5/23	17/23
[0.7, 0.9)	7/32	29/32
[0.9, 1.0]	12/36	33/36
Novel controls [†]	6/28	7/28

[†] The Novel queries and labels are retained from the main evaluation.

TABLE VI

REAL-WORLD REVISIT NAVIGATION. SUCCESSES/TRIALS ACROSS EIGHT INDOOR AND TWO OUTDOOR SETTINGS.

Setting	Target landmark	Base	GEM
Indoor 1	Stairwell atrium	1/3	2/3
Indoor 2	Corridor door, cart	0/3	3/3
Indoor 3	Corridor, fan and boxes	1/3	2/3
Indoor 4	Corridor door, notices	1/3	3/3
Indoor 5	Projection screen	1/3	2/3
Indoor 6	Glazed office bay	0/3	2/3
Indoor 7	Carpeted store bay	0/3	2/3
Indoor 8	Lift lobby, planter	0/3	3/3
Outdoor 1	Glazed entrance	0/3	3/3
Outdoor 2	Roof terrace, benches	0/3	3/3
Overall		4/30	25/30

key/value (KV) storage, compact and full depth-map archives yield identical actions on 140 paired queries from 70 histories. On 234- and 1,948-frame histories, compact storage reduces historical depth/confidence bytes by $39.74\times$ and $43.58\times$, excluding RGB, descriptors, and neural state.

Optional lossless encoding compresses recent-view KV tensors and restores their original BF16 values for attention. In a separate 1,948-frame run, it lowers the geometry module’s peak GPU memory from 13.12 to 11.40 GiB, with matching geometry outputs, archived records, and final Novel/Revisit readouts.

Deployment latency. A stationary Jetson–RTX 4090 replay uses a 120-frame real survey and a dedicated GPU. Across 121 planning calls after 15 warmup calls, median/P95 request-to-response latency is 387.7/490.8 ms with full depth maps and 336.0/394.1 ms with compact archives. Adding lossless KV encoding gives 334.4/393.7 ms. Timing includes transmission of prepared JPEGs, geometry updates and goal readout, NavDP inference, and response delivery. All three configurations use BF16 readout and return identical depth, bearings, and trajectories.

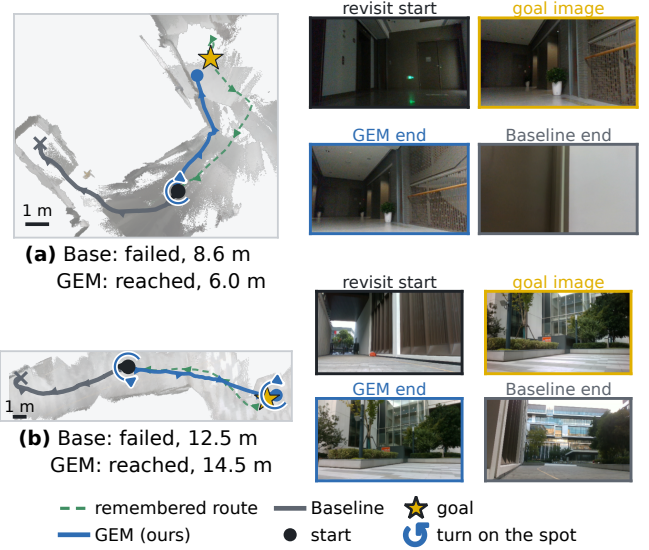


Fig. 3. **Real-world Revisit trials.** Indoor (a) and outdoor (b) pairs share the goal image, recorded survey, and start pose facing away from the goal. GEM reads the survey history (dashed green) to recover the goal direction. Circular arrows mark turns in place; map overlays are for visualization.

H. Real-World Evaluation

Platform and paired protocol. We use a Unitree Go2 with a RealSense D435i. An onboard Jetson handles image capture and execution; policy inference runs on an external RTX 4090. Both methods run frozen NavDP on RGB and predicted monocular depth. GEM also uses RGB history to find the goal direction. Sensor depth is used only by the local safety layer.

We evaluate 30 trials per method in eight indoor and two outdoor settings (Table VI). Paired runs share the starting pose, goal image, and teleoperated survey. Before each run, we restore the surveyed geometric memory and initialize NavDP’s observation buffer with the current live view. Subsequent RGB observations extend the restored geometry stream. An RGB homography check triggers automatic stopping; we assess goal arrival by on-site visual inspection. A run fails if it times out or the robot leaves the traversable area.

Open-space revisits. GEM reaches the goal in 25/30 trials, compared with 4/30 for Base (Table VI). Figure 3 shows indoor and outdoor cases where GEM recalls the surveyed goal and guides the frozen controller back to it.

VI. LIMITATIONS

Geometric support and long-range navigation. GEM requires verifiable historical overlap, and performance degrades as support weakens (Table V). Verification checks goal-to-history consistency, while history-to-current pose drift can still degrade the recovered bearing. Even an accurate bearing does not encode obstacle topology or a traversable route. Over longer distances, passages and detours can require sustained motion away from the goal. Extending recall to such routes calls for route reasoning beyond the current local cue.

Memory and computational cost. The recent dense-view window is limited to 64 committed views, while historical trajectory tokens and archived observations continue to grow with episode length. The compact archive reduces historical depth and confidence storage, and lossless encoding reduces the working-state footprint within the same context, but neither bounds memory growth over arbitrarily long episodes. Bounded memory with predictable long-horizon latency therefore remains an open problem.

VII. CONCLUSION

GEM lets frozen navigation policies reuse past observations by pairing streaming geometric state with an indexed image archive. A verified historical view links the goal to the current camera, while the same stream supplies monocular depth. Across NavDP, ViNT, and NoMaD on shared causal histories in HM3D and MP3D, GEM improves Revisit success by 60.7–81.0 percentage points while maintaining Novel performance. The same memory supports successive goals in simulation and Revisit navigation on a real robot. Together, these results show how geometry accumulated during navigation can make past visual experience useful to a frozen controller.

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